

407/60

END-SEMESTER EXAMINATION, May-2025
Computer Networking: Security (CSE 3752)

Program: B.Tech (CSE/CSIT)
Full Marks: 60

Semester: 6th
Time: 3 Hours

- (b) Name and define the method that limits the impact of breach. 2
- (c) List the different ways by which the cyber intruders are prevented from moving freely across the entire network. 2
- 9 (a) List and highlight the use of key elements for a multilayer strategy required to be implemented towards achieving effective network security. 2
- (b) Explain briefly the micro-services as a secured architecture listing out the key benefits of it. 2
- (c) With the help of suitable diagram, explain the function of each of the four Supervisory Control and Data Acquisition (SCADA) system levels. 2
- 10 (a) With the help of suitable examples highlight the attributes associated with different categories of devices in the context of infrastructure security. 2
- (b) List and define the various data types considered as a regulated data required to be protected from security breach in organizations. 2
- (c) A company notices that one of its web servers is receiving significantly more traffic than others, leading to performance issues. To overcome the issue the network engineer uses the load balancer to distribute the traffic among the web servers on rotational basis. Name and explain the scheduling technique used by the load balancer that best suits the given scenario. 2

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Understand the various types of security controls and fundamental security concepts.	L1	1	6
Explain the importance of using appropriate cryptographic solutions like PKI, encryption levels, tools, obfuscation, hashing, digital signatures, and certificates.	L3	2,3,4	18
Explain common threat actors, their motivations, and various threat vectors, along with understanding different types of vulnerabilities.	L2	5,6	12
Analyze the indicators of malicious activity, based on a given scenario and explain the purpose of mitigation techniques used to secure the enterprise.	L2	7,8	12
Explain the importance of the security implications of different architecture models, including cloud, serverless, microservices, and network infrastructure.	L2	9	6
Apply security principles in different infrastructure scenarios including device placement, security zones, and network appliances.	L2	10	6

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

- 1. (a) List out the common control categories associated with security management of an organization. With the help of suitable examples briefly explain, how technical control improves the security performance. 2

- (b) Discuss the significance of non-repudiation by highlighting its key aspects with respect to cyber communication. 2
- (c) Briefly illustrate the following factors that has a noticeable impact on change management. 2
- (i) Downtime (ii) Service restart (iii) Application restart
(iv) Legacy applications
2. (a) With the help of suitable examples differentiate between substitution and transposition based symmetric cryptography. 2
- (b) State the significance of the two different keys used in public key cryptography. Mention, how these two keys are used for the following two scenarios using public key infrastructure: 2
- (a) Bob wants to send message to Alice securely.
- (b) Bob wants to make sure that the message received by him is from Alice only.
- (c) Encrypt the plaintext "EXAMINATION" using the keyword "LEG" with the help of Vigenère cipher. 2
3. (a) State, how both substitution and permutation operations are being a part of DES encryption technique which makes it quite difficult to cryptanalyze. 2
- (b) State the significance of extension permutation table in DES algorithm. Given a 32-bit input as (1234ABCD)_{Hex} to the expansion permutation table used in DES algorithm. Find the respective 48 bit output (in Hexadecimal form). 2
- (c) The 56 bit output of permutation choice 1 table for generating the round keys in DES is (01020304050607)_{Hex}. Find the output (in Hexadecimal form) after circular left shift operation if the same is used to generate the sub key for round 1. 2
4. (a) List out the similarity and differences between AES and DES encryption process. 2
- (b) Given the content of state array after initial round key operation in AES 128 encryption process as (000102030405060708090A0B0C0D0E0F)_{Hex}. 2
- (i) Show the content of output state array (in 4 X 4 matrix form) after Sub byte transformation.
- (ii) Show the content of output state array (in 4 X 4 matrix form) after Shift row transformation.
- Assume the elements of 1st row (i.e. row no. 0) in AES S-box as 63, 7C, 77, 7B, F2, 6B, 6F, C5, 30, 01, 67, 2B, FE, D7, A8, 76.

- (c) Joy wants to send the message $M = 13$ to Mike privately, which can only be decrypted by Mike. Which key of Mike, Joy will use to encrypt its message. If Mike's private key is (33, 7) and public key is (33, 3), find the value of encrypted message C. 2
5. (a) A former employee, who is terminated, hacks into a company's database to delete critical customer record to disrupt business operation because of lasting grievance around their termination. Explain the type of motivation that is reflected in the scenario. 2
- (b) List the distinguished feature of a business email compromise attack. 2
- (c) What is meant by vulnerable software and explain how the vulnerabilities occurs. 2
6. (a) Describe a memory injection attack and the risks it poses to application security. 2
- (b) What is meant by Structured Query Language Injection (SQLi) attack? Explain the steps used in this attack to make a system infectious. 2
- (c) List and define the flaws in cryptography process which may lead to a possible cryptographic vulnerability. 2
7. (a) What is meant by DNS cache poisoning associated with DNS attack? Discuss briefly, how DNS sinkhole can be used to prevent DNS attack. 2
- (b) During assessment of security it is noticed that on Monday morning at 9 am, the files of a company's Chief Financial Officer (CFO) are deleted without any warning. The IT Support team restored the data, but on the following Monday morning at 9 am, the files were again deleted. Name and explain the type of attack described by the scenario. 2
- (c) State the essence of indicators of attack as far as the information security in computer network is concerned. Discuss briefly how a blocked content can be used as an indicator of attack. 2
8. (a) In a large enterprise network, the human resources department and the IT department each require isolation from the rest of the company's network. What security technique is adopted to achieve the isolation and why. 2

END-SEMESTER EXAMINATION, May-2025
Essential Mathematics for Artificial Intelligence 2 (CSE 3788)

Programme: B.Tech(CSE & CSIT)
 Full Marks: 60

Semester: 6th
 Time: 3 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Able to understand about different types of probabilistic generative model and use of it in probabilistic language modelling.	L3, L3, L5, L3	1(a,b,c) 3(a)	2*4=8
Able to apply the knowledge of graph theory for solving different AI model.	L3, L3, L3, L3, L4, L3	2(a,b,c), 3(b),5(b,c)	2*6=12
Able to apply machine learning into the field of operations research for more efficient and informed decision making.	L3, L3, L3, L3, L3, L3	3(c),4(a) 4(b,c), 5(a), 6(b)	2*6=12
Able to analyse the use of different stochastic method for solving AI model.	L3, L3, L1, L3, L3	6(a,c) 7(a,b,c)	2*5=10
Able to use the concept of mathematical logic and partial differential equation in the field of AI.	L1, L3, L3, L4, L3, L3, L3	8(a,b,c), 9(a,b,c), 10(a)	2*7=14
Able to understand about AI ethics and AI policy.	L1, L3	10(b,c)	2*2=4

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1. (a) Calculate the probability of randomly selecting 4 good quarts of milk in succession from a cooler containing 20 quarts, of which 5 have spoiled. 2
- (b) Determine the Empirical distribution $\hat{F}_n(x)$ of the data [0, 1, 2, 2, 4, 6, 6, 7]. Also, draw the graph of $\hat{F}_n(x)$. 2
- (c) Consider a sample consisting of the values 12, 11.2, 13.5, 12.3, 13.8, and 11.9 comes from a population with the density function 2

$$f(x, \theta) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & x > 1 \\ 0 & \text{elsewhere} \end{cases}$$

where $\theta > 0$. Evaluate the maximum likelihood estimate of θ .

7. (a) What is transient state and recurrent state? Give example using a transition diagram of a Markov chain. 2
- (b) Consider an MDP with 3 states $S=\{S_1, S_2, S_3\}$ and 2 possible actions $A=\{A_1, A_2\}$. The transition probabilities P and rewards R for each state-action pair are given as follows:
 $P(S_2|S_1, A_1)=1, R(S_1, A_1)=5; P(S_1|S_1, A_2)=1, R(S_1, A_2)=0;$
 $P(S_3|S_2, A_1)=1, R(S_2, A_1)=10; P(S_1|S_2, A_2)=1, R(S_2, A_2)=2;$
 $P(S_4|S_3, A_1)=1, R(S_3, A_1)=2; P(S_3|S_3, A_2)=1, R(S_3, A_2)=0.$
 Discount factor $\gamma=0.9$ and initial values of all states are 0. Use Value Iteration to calculate the values of the states S_1, S_2 and S_3 after 1st iteration. 2
- (c) Discuss which action is best for S_1, S_2 and S_3 after 1st iteration. 2
8. (a) What is the difference between fuzzy logic & probability? 2
- (b) Let $P(x)$ be the statement "x spends more than 7 hours every weekday in office", where the domain for x consists of all employees. Demonstrate each of these in English:
 i) $\forall x \neg P(x)$,
 ii) $\exists x \neg P(x)$. 2
- (c) Express the statement "Some student in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using predicates & quantifiers. 2
9. (a) Differentiate between ordinary differential equation and partial differential equation. 2
- (b) Solve $\frac{d^2 y}{dx^2} = 1$ on $[0,1]$ with boundary conditions $y(0)=-1, y(1)=0$ by analytical method. 2
- (c) Solve the above ODE by finite difference method when the interval is divided into 3 equally spaced points. 2
10. (a) Derive the discretization of the one-dimensional heat equation $u_t = \alpha u_{xx}$. 2
- (b) What is Good AI? Give example. 2
- (c) Discuss about AI policies and its type. 2

End of Questions

Consider a small web network with four pages: A, B, C, and D. The link structure is as follows: Page A links to B and D, Page B links to A, Page C links to B, and Page D links to C and A. Create the graph.

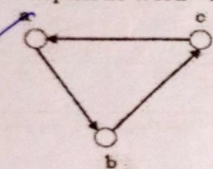
(b) Construct the transition probability matrix for the aforementioned web network.

(c) Calculate the page rank for all pages of the given network up to one iteration after initial iteration when the damping factor is 0.85.

3. (a) Consider a dataset of 6 emails detected as Spam(S) or Not Spam (NS), based on the features: Sender Type: Known (K) or Unknown (U) and Suspicious Word Present: Yes (Y) or No (N).

Email	Sender Type	Suspicious Word	Class
E1	K	N	NS
E2	U	Y	S
E3	U	Y	S
E4	K	N	NS
E5	U	N	NS
E6	K	Y	S

Use Naïve Bayes Classification to determine whether an email is Spam(S) or Not Spam (NS) if "Sender = Unknown (U) and Suspicious Word = Yes (Y)".



Let $a[1, 0, 1]$, $b[0, 2, 2]$ and $c[3, 1, 2]$ be three nodes of this graph. Then calculate the updated nodes after message passing where aggregated message will be the cross product of nodes.

(c) Use graphical method to solve the following Linear programming problem and also indicate the feasible region

$$\min Z = 12x_1 + 16x_2 \quad \text{Subject to } x_1 + 2x_2 \geq 30, x_1 + x_2 \geq 20, x_1, x_2 \geq 0.$$

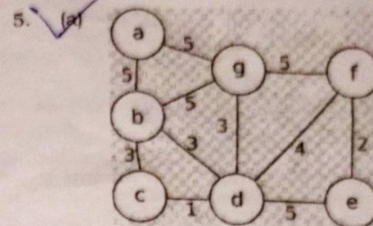
4. (a) Using simplex method write the initial table for the following LPP:

$$\max Z = 40x_1 + 30x_2 \quad \text{Subject to } x_1 + x_2 \leq 12$$

$$2x_1 + x_2 \leq 16, x_1, x_2 \geq 0.$$

(b) Calculate the first iteration table for the above LPP.

(c) Derive the second iteration table and obtain the solution, maximum value of Z for the LPP given in 4(a).



Construct the minimum spanning tree using Prim's algorithm by taking the starting node as 'a'.

(b) A doctor is diagnosing whether a patient has lung cancer (C) based on two symptoms: cough (S1) and fatigue (S2). The presence of lung cancer is influenced by whether the patient smokes (R). The Bayesian network structure is as follows: Smoking (R) affects Lung Cancer (C), Lung Cancer (C) affects Cough (S1) and Fatigue (S2). Generate the directed acyclic graph (DAG).

(c) Given the following probabilities for the aforementioned DAG

$$P(R=1)=0.3, P(C=1|R=1)=0.4, P(C=1|R=0)=0.05,$$

$$P(S_1=1|C=1)=0.8, P(S_1=1|C=0)=0.3, P(S_2=1|C=1)=0.7,$$

$$P(S_2=1|C=0)=0.2, \text{ then calculate } P(S_1, C, R).$$

6. (a) Three prisoners, A, B, and C, are in separate cells and sentenced to death. The governor has selected one of them at random to be pardoned. The warden knows which one is pardoned, but is not allowed to tell. Prisoner A begs the warden to let him know the identity of one of the two who are going to be executed. If B is to be pardoned, give me C's name. If C is to be pardoned, give me B's name. And if I'm to be pardoned, secretly flip a coin to decide whether to give me the name B or C. The warden gives him B's name. Calculate the probability of A's chance of being pardoned after knowing the Warden's information

(b) Calculate the value of x, y by using Lagrange multiplier technique to Maximize $z = x^{0.2}y^{0.4}$, subject to $50x + 5y \leq 5000$.

(c) A meteorologist is modeling the weather forecast for a city. The weather can be in one of two states: Sunny (State 1), Rainy (State 2). The transition probabilities between these two states are as follows: If it is sunny today, the probability that it will be sunny tomorrow is 0.7, and the probability that it will be rainy tomorrow is 0.3. If it is rainy today, the probability that it will be sunny tomorrow is 0.40, and the probability that it will be rainy tomorrow is 0.6. The meteorologist knows that today, the weather is sunny. Derive probability that the weather will be rainy two days from now?

END-SEMESTER EXAMINATION, May-2025

Compilers: Principles, Techniques, and Tool (CSE 3739)

Programme: B.Tech(CSE,CSIT)
Full Marks: 60

Semester: 6th
Time: 3 Hours

		Here # and % are operators and id is a token that represents an integer and id.val represents the corresponding integer value. Draw the annotated parse tree for the input expression: 20#10%5#8%2%2 and compute the final value of S.val.	
	(b)	Given the following syntax-directed translation rules: Rule 1: $R \rightarrow AB \{ B.i = R.i - 1; A.i = B.i; R.i = A.i + 1; \}$ Rule 2: $P \rightarrow CD \{ P.i = C.i + D.i; D.i = C.i + 2; \}$ Rule 3: $Q \rightarrow EF \{ Q.i = E.i + F.i; \}$ Assume all attributes are integers. For each rule, identify which attributes are inherited and which are synthesized.	2
	(c)	Consider the following intermediate code segment: $x = u - t;$ $y = x * v;$ $x = y + w;$ $y = t - z;$ $y = x * y;$ Construct the Directed Acyclic Graph (DAG) representation for the above code segment.	2
10	(a)	L1: $t1 = -1$ L2: $t2 = 0$ L3: $t3 = 0$ L4: $t4 = 4 * t3$ L5: $t5 = 4 * t2$ L6: $t6 = t5 * M$ L7: $t7 = t4 + t6$ L8: $t8 = a[t7]$ L9: if $t8 \leq \text{max}$ goto L11 L10: $t1 = t8$ L11: $t3 = t3 + 1$ L12: if $t3 < M$ goto L4 L13: $t2 = t2 + 1$ L14: if $t2 < N$ goto L3 L15: $\text{max} = t1$ Divide the given pseudo-code into basic blocks. Then, determine the number of instructions in the largest basic block.	2
	(b)	Generate the assembly language code for the given intermediate code: $t1 = a + b$ $t2 = c + d$ $t3 = e - t2$ $t4 = t1 - t3$	2
	(c)	Discuss various Loop optimization techniques with example.	2
		End of Questions	

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Understand the overview of programming languages, language processors and the structure of a compiler.	L3, L4	1,2	12
Acquire knowledge in theory of computation and their role in designing different types of tokens generated by lexical analyzer.	L3	3,4	12
Understand the role of Parser(s) (LL, SLR, CLR and LALR) and its types i.e. Top-down and Bottom-up parsers.	L6	5,6,7,8	24
Apply and evaluate syntax directed translation schemes, synthesized attributes, inherited attributes, and different techniques for symbol table organization	L5	9	6
Analyze the generation of various intermediate codes and the process of their optimization.	L4	10. a,c	4
Understand the target machine's run time environment, and its instruction set for code generation and techniques used for code optimization	L3	10. b	2

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1.	(a)	Write the output of 1 st four phases of the compiler for the following arithmetic expression $x = (a+b) * -c/d$	2
	(b)	Consider the following fragment of 'C' code: float a,b; a = a * -50 + b + 3; Write the output of last 3-phases of the compiler for the above arithmetic expression in the 'C' code.	2
	(c)	Describe the importance of symbol table during compilation process of a program.	2

2.	(a)	Provide a comparative analysis of a compiler and an interpreter, highlighting their differences and similarities.	2
	(b)	Define a hybrid compiler and explain its working with the help of a diagram.	2
	(c)	Consider the following C program: <pre>#define a (x+1) int x = 2; void b() { x = a; printf(" %d\n", x); } void c() { int x = 1; printf(" %d\n", a); } void main() { b(); c(); }</pre> Identify the output of the program and explain how the macro #define a (x+1), affects the scope of the variables a and x in the functions b() and c() .	2
3.	(a)	Define Token, Lexeme and Pattern with suitable examples.	2
	(b)	Write regular definitions for the valid lexical structure of floating-point constants in the C programming language.	2
	(c)	The pattern for the token relop is defined as: relop $\rightarrow < > < = > = = < >$ Construct the state transition diagram that will recognize the token relop .	2
4.	(a)	What are lexical errors in the context of lexical analysis? Describe how panic mode recovery is used to handle such errors.	2
	(b)	Write regular definitions for the valid lexical structure of identifiers in the C programming language.	2
	(c)	Consider the following C program, identify the tokens, and count the total number of tokens. <pre>int main(i, j) { int i, j; return i*j? i : j; }</pre>	2
5.	(a)	Consider the language L defined by the regular expression $(a b)^*a(a b)^*$ over the alphabet $\{a, b\}$. Construct an NFA that accepts this language.	2
	(b)	Construct the equivalent DFA from the NFA constructed in Question 5(a).	2
	(c)	Write the regular definition for natural numbers that is divisible by 5. A number is considered divisible by 5 if its last digit is either 0 or 5. Examples of such numbers include 5280, 115, 105, 5, and 2025	2
6.	(a)	Check the given CFG is having common prefix or not. If yes, remove it from the grammar and rewrite the grammar. $E \rightarrow iEtS iEtMeS c$ $M \rightarrow b$	2

	(b)	Consider the Context-free Grammar: $S \rightarrow SS^+ \mid SS^* \mid a$ Give a rightmost derivation and construct the parse tree for the string $w = aa + a^*$	2															
	(c)	Let G be the following context-free grammar, where E is the start symbol of G ; $E \rightarrow E \# E \mid E \$ E \mid E ? E \mid E @ E \mid id$ Construct an unambiguous grammar G' for above ambiguous grammar G with reference to the operator table given below. [Hint: Assume #, \$, ? and @ are binary operators]	2															
		<table border="1"><tr><th>operator</th><th>Associativity</th><th>precedence</th></tr><tr><td>@</td><td>right to left</td><td>lowest</td></tr><tr><td>#</td><td>right to left</td><td></td></tr><tr><td>?</td><td>left to right</td><td></td></tr><tr><td>\$</td><td>left to right</td><td>highest</td></tr></table>	operator	Associativity	precedence	@	right to left	lowest	#	right to left		?	left to right		\$	left to right	highest	
operator	Associativity	precedence																
@	right to left	lowest																
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7.	(a)	Given the context-free grammar with its productions: $S \rightarrow S(S)S \mid \epsilon$ Show that it is not suitable for a top-down predictive parser. Identify the problem and correct it by rewriting the grammar.	2															
	(b)	Consider the following context-free grammar: $S \rightarrow AA$ $A \rightarrow aA \mid b$ Construct the canonical collection of LR(0) items for this grammar and draw the LR(0) transition diagram (DFA of items).	2															
	(c)	Construct the LR(0) parsing table for the grammar given in Question 7(b) and demonstrate the parsing actions (shift, reduce, accept, error) for the input string: aabb	2															
8.	(a)	Define handle pruning in the context of bottom-up parsing. Consider the following grammar: $E \rightarrow E + n \mid E \times n \mid n$ Determine the handles and Construct the bottom-up (shift-reduce) parse for the string: $n + n \times n$	2															
	(b)	Consider the following context-free grammar, $E \rightarrow E - T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow a$ Find the canonical collection of LR(1) sets of items and construct the transition diagram.	2															
	(c)	Construct the CLR(1) parsing table for the Grammar defined in Question 8(b).	2															
9.	(a)	Consider the following grammar with syntax-directed translation rules: 1) $S \rightarrow S \# T$ { $S.val = S.val * T.val$ } 2) $S \rightarrow T$ { $S.val = T.val$ } 3) $T \rightarrow T \% R$ { $T.val = T.val + R.val$ } 4) $T \rightarrow R$ { $T.val = R.val$ } 5) $R \rightarrow id$ { $R.val = id.val$ }	2															

END-SEMESTER EXAMINATION, May-2025 Introduction to databases (CSE-3151)

Programme: B.Tech (ALL CSE & CSIT)
Full Marks: 60

Semester: 6th
Time: 3 Hours

9. (a) Prove or disprove: Every recoverable schedule is cascadeless. 2
Justify your answer with an example.
- (b) Consider schedule S with transaction T1 and T2. T1 transfer Rs.500 from account A to B, and T2 adds Rs.100 into account A. Prepare concurrent schedule with two phase locking protocol. 2
- (c) The transactions T1, T2 with time stamps (TS) 5, 10 respectively want to write data item X. The current read time stamp is RTS(X) = 8 and write time stamp is WTS(X) = 7. Will the write be allowed for both the transactions T1, T2? Justify your answer. 2
- 10 (a) Given the following transactions and their timestamps, simulate the timestamp ordering protocol and determine whether each operation is allowed or leads to a rollback. Initial: RTS(X)=0, WTS(X)=0 2

T1: TS = 10	T2: TS = 20
Read(X)	Read(X)
Write(X)	Write(X)

- (b) Write down two major differences between immediate and deferred database modification. 2
- (c) What actions will be taken by the log based recovery technique for deferred and immediate database modification, if a failure occurs after the Write(B) operation in the following schedule: 2

T1:	T2:	T3:
Read(A) A=A-500	Read(B) B=B+100	
Write(A) Commit	Write(B) Commit	Read(C)

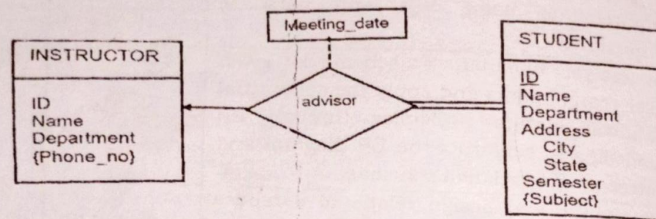
End of Questions

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
To identify and explain the different components and functionalities of DBMS and their interdependence through the database architecture.	L1, L2	1a,b,c	6
To formulate SQL queries and able to develop a database application as per user requirements using SQL and database connectivity.	L3	7b	2
To analyze an enterprise schema for given user requirements and apply the conceptual database design principles through ER modelling to construct the ER diagram and corresponding relation schemas.	L2, L4	2a,c 3a,b,c	10
To analyze and design relational database schema using decomposition and normalization techniques.	L2,L5, L6	2b 4a,b,c 5a,b,c	14
To apply relational algebra and relational calculus to express queries on relational schemas.	L4	6a,b,c 7a,c	10
To interpret the functional issues related to transaction and database recovery along with the concept of storage and database system architecture.	L4	8a,b,c 9a,b,c 10a,b,c	18

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

- Answer all questions. Each question carries equal mark.
1. (a) What is a data model? Differentiate between relational and semi-structure data model. 2
- (b) Explain the roles of the query processor and the storage manager in database architecture. 2
- (c) How DDL and DML statements affect instances and schemas of a database? 2

- (a) Discuss the binary and ternary degree of relationships in the ER model with suitable examples.
- (b) Consider a relational schema $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of functional dependencies $EF \rightarrow G, F \rightarrow IJ, EH \rightarrow KL, K \rightarrow M, L \rightarrow N$ on R . Find the candidate key and at least one super key of R . (GATE 2014)
- (c) Distinguish between attribute inheritance and participation inheritance in specialization with example.
3. (a) An entity set order with attributes order_num, order_date and payment_info, and another entity set order_item with attribute item_num, quantity and price are associated with a relationship set "contains". The existence of order_item is dependent on the order. Construct the suitable ER Diagram to represent this scenario.
- (b) Convert the given part of ER diagram into minimum number of relational schemas.



- (a) Analyze the relational schema given in 3(b), and draw its corresponding schema diagram.
4. (a) Explain First Normal Form (1NF) with a suitable example. Why 1NF is considered a foundational step in normalization?
- (b) Find the canonical cover of the following set F of functional dependencies on the schema $R(A, B, C)$, $F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$.
- (c) What is partial dependency? Identify the number of partial dependencies in the relational schema $R(A, B, C, D, E)$ with FD set $F = \{A \rightarrow BE, E \rightarrow C, C \rightarrow AB\}$.
5. (a) Consider the relation schema $R(A, B, C, D, E)$ with functional dependency set $F = \{A \rightarrow BE, E \rightarrow C, C \rightarrow AB\}$. Comment whether it satisfies 2NF or not with reason. If no, redesign the schema satisfying the properties of 2NF.
- (b) Consider the relational schema Student_Performance(Name, CourseNo, RegNo, Grade) has the following functional dependencies: $F = \{(Name, CourseNo) \rightarrow Grade, (RegNo, CourseNo) \rightarrow Grade, Name \rightarrow RegNo, RegNo \rightarrow Name\}$. Comment whether it satisfies BCNF or not with reason. If no, redesign the schema satisfying the properties of BCNF. (GATE 2019)

- (c) Consider the relation schema $R(A, B, C, D)$ with functional dependency set $F = \{A \rightarrow D, A \rightarrow B, A \rightarrow C\}$. Comment whether it satisfies 4NF or not with reason. If no, redesign the schema satisfying the properties of 4NF.
6. (a) Consider the following relational schema, employee (person-name, street, city), works (person-name, company-name, salary), Company (company-name, city), where the primary keys are underlined. Write a relational algebra query to find the names, street address, and cities of residence of all employees who work for TCS and earn more than 40,000 per annum.
- (b) Write a query using Tuple relational calculus to find the name and company name of the employee who live in the same city as the company for which they work.
- (c) Write a query using Domain relational calculus to display the name and salary of the employees working in the company placed in Delhi and employees belong to Mumbai.
7. (a) Assume that the companies may be located in several cities. Write a relational algebra query to find all company name located in every city where Infosys is located. (consider the schema given in 6(a))
- (b) Consider two relations $R_1(A, B, D)$ with tuples $(1, 5, 2), (3, 7, 4)$, and $R_2(A, C) = (1, 7), (4, 9)$. Find out the left and right outer join of R_1 and R_2 .
- (c) Consider the following relation R

A	B	C	D
a	a	10	7
a	b	15	7
b	b	12	5
b	b	20	10

Write down the output of the following command.

- i) $\Pi_{A,B}(\sigma_{C > 10}(R))$ ii) $\Pi_{A,B}(\sigma_{A=B \wedge D \geq 5}(R))$
8. (a) Explain how each of the ACID properties ensures reliable transaction processing in DBMS.
- (b) Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data item z by a transaction T_i , respectively. Check whether the given schedule S with three transactions is a conflict serializable schedule or not.
 $S: R_1(x) R_2(x) W_1(x) R_3(x) W_2(x)$.
- (c) Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data item z by a transaction T_i , respectively. Check whether the given schedule S_1 with four transactions is conflict serializable and recoverable or not. Justify your answer.
 $S_1: R_2(A) W_3(A) \text{Commit}(T_3) W_1(A) \text{Commit}(T_1) W_2(B) R_2(C) \text{Commit}(T_2) R_4(A) R_4(B) \text{Commit}(T_4)$. (GATE 2014)

8. (a) Explain the difference between simple linear regression and multiple linear regression. What is the utility of using multiple linear regression in real-world datasets like the **California Housing dataset**? 2
- (b) Develop a Python code that loads the **California Housing dataset**, splits it, trains a multiple linear regression model, and outputs **R²** and **MSE** scores. 2
- (c) What is multicollinearity, and how can it affect the interpretation of coefficients in a multiple linear regression model such as the one built on the **California Housing dataset**? 2
9. (a) What are the key differences between classification and clustering? How is **K-Means clustering** different from **KNN**? 2
- (b) Answer the following questions: 2
- Can dimensionality reduction lead to a loss of information? How can this be minimized?
 - In what situations would you prefer **t-SNE** over **PCA**?
- (c) Use the **Iris** dataset to perform unsupervised learning with the **K-Means clustering** algorithm in Python, and evaluate how well the clusters align with the actual species. 2
10. (a) What is the primary purpose of an activation function in a neural network? 2
- (b) Write a Python program using **Keras** to train an **Artificial Neural Network (ANN)** on the following dataset to classify students as Pass or Fail. Here, x_1 represents the number of hours studied, x_2 represents the attendance percentage, and y is the target label where 1 = Pass and 0 = Fail. After training, predict the outcome for a student who studied 6 hours and had 90% attendance. 2
- Dataset:**
- | Hours Studied (x_1) | Attendance (%) (x_2) | Pass/Fail (y) |
|-------------------------|--------------------------|-------------------|
| 6 | 90 | 1 |
| 3 | 70 | 0 |
| 8 | 95 | 1 |
| 4 | 65 | 0 |
| 10 | 80 | 1 |
| 5 | 85 | 1 |
- (c) Describe the key layers of a **Convolutional Neural Network (CNN)** and explain their roles in performing image classification. 2

End of Questions

END-SEMESTER EXAMINATION, May-2025
Python for Computer Science and Data Science 2
(CSE 3652)

Programme: B.Tech (CSE)
 Full Marks: 60

Semester: 6th
 Time: 3 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Apply object-oriented programming principles and advanced Python features to design and test reusable, efficient applications for real-world problems.	L1, L2, L3	1(a, b, c) 2(a)	8
Analyze and implement recursive algorithms, searching, and sorting techniques, and evaluate their efficiency using Big O notation to optimize performance.	L3, L4	2 (b, c) 3(a)	6
Apply natural language processing (NLP) techniques using tools and libraries to analyze, process, and visualize text data, including sentiment analysis, language translation, and entity recognition, with applications in machine learning.	L3, L4	3 (b, c) 4(a, b, c) 5(a)	12
Utilize APIs to collect, analyze, and visualize social media data, identify trends, and develop cognitive computing applications.	L3, L4	5(b, c)	4
Apply supervised and unsupervised machine learning techniques, including classification, regression, clustering, and dimensionality reduction, using libraries and APIs to analyze data, evaluate models, and solve real-world problems.	L3, L4	6(a, b, c) 7(a, b, c) 8(a, b, c) 9(a, b, c)	24
Implement deep learning models for applications like image classification, sentiment analysis, and reinforcement learning, while using big data technologies to process and visualize large datasets efficiently.	L3, L4	10(a, b, c)	6

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

- Answer all questions. Each question carries an equal mark.
1. (a) What is the purpose of the **`__init__`** method in Python? 2
- (b) Write a Python program to overload the **`+`** operator to perform the addition of two complex numbers using a custom Complex class. 2
- (c) Write a Python program demonstrating polymorphism by creating a base class **`Shape`** with a method **`area()`**, and two subclasses **`Circle`** and **`Rectangle`** that override the **`area()`** method. Instantiate objects of both subclasses and call the **`area()`** method. 2

What will be the output for the following Python code?

```
class MyClass:
    def __init__(self):
        self.__hidden = "Private Attribute"

    def reveal(self):
        return self.__hidden
```

```
obj = MyClass()
print(obj.reveal())
print(obj.__MyClass__hidden)
```

- (b) Compare recursion and iteration in terms of memory usage and performance. Illustrate your answer with the example of computing the factorial of a number.
- (c) Write a Python program to implement the **Binary Search** algorithm and compare its time complexity with that of **Linear Search**. Under what conditions is Binary Search preferred over Linear Search?
3. (a) Explain the **Merge Sort** algorithm. Provide a detailed analysis of its time and space complexities in the best, average, and worst-case scenarios.
- (b) What is the **NaiveBayesAnalyzer** in TextBlob, and how does it differ from the default sentiment analysis method used in TextBlob?
- (c) Write a Python program using **TextBlob** to tokenize a given paragraph into sentences and words. Display each sentence and the list of words it contains (excluding punctuation).
4. (a) Write a Python program using **TextBlob** to convert the singular word 'index' to its plural form, convert the plural word 'cacti' to its singular form, and convert the words 'cat dog rat' into their plural forms using a **WordList**, and display the results of each operation.
- (b) Write a Python program to extract all synonyms and antonyms of the word 'happy' using **WordNet** (via **TextBlob** or **NLTK**).
- (c) In the context of text preprocessing, how do stemming and lemmatization affect the analysis of word occurrences in a given corpus? Explain with an example.
5. (a) Write a Python program to remove English stop words from the sentence "Today is a beautiful day." using **NLTK**, and then generate trigrams from the sentence "Today is a beautiful day: Tomorrow looks like bad weather." using **TextBlob**. Display the results.
- (b) What is IBM Watson, and what are some of the capabilities it offers through its cloud-based platform?
- (c) Write a Python program to search for recent tweets containing a specific keyword using the **Tweepy** package and authenticate with the Twitter API.

6. (a) Write a Python expression to split a dataset into training and testing sets (80% training, 20% testing), where **X** represents the features and **y** represents the target labels.
- (b) How can **cross-validation** help in evaluating a classification model? Observe the code snippet provided below:

```
from sklearn.model_selection import KFold
kfold = KFold(n_splits=10, random_state=11,
              shuffle=True)
```

Assuming you are applying **k-fold** on a dataset with 100 data points, explain the terms in the code snippet.

- (c) A confusion matrix is given below:

Prediction ↓	Actual →	Actual Apple	Actual Banana	Actual Cherry
Predict Apple		1	2	1
Predict Banana		0	2	1
Predict Cherry		2	0	1

Calculate the **True Positive**, **False positive**, **False Negative**, and **True Negative** from the above table. Also, write the accuracies of Apple, Banana, and Cherry.

7. (a) What is **underfitting** in the context of simple linear regression? How can it affect model performance?
- (b) You are given the following dataset, where **x** represents the number of hours studied and **y** represents the exam score.

Hours Study(x)	Exam Score(y)
1	50
2	55
3	60
4	65

Write a Python program to implement (**only fit**) a simple linear model for this data.

- (c) Consider the following data points and the regression equation $Y=2X+3$.

Input(X)	Output(Y)
1	5
2	8
3	11
4	14

Calculate the **Mean Squared Error (MSE)** for this regression model and explain it.

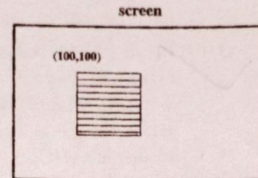
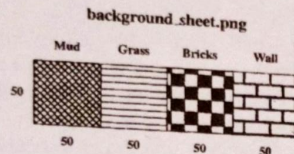
END-SEMESTER EXAMINATION, May-2025

Game Programming with C++ (CSE 3545)

Programme: B.Tech.(CSE)
Full marks: 60

Semester: 6th
Time: 3 hours

- (b) Using SFML and C++, write the code to update the position of a zombie so that it moves toward the player. Assume the zombie's current position is (x_2, y_2) and the player's position is (x_1, y_1) . [2]
- (c) Write a C++ SFML code snippet to create 5 zombies placed at random positions within an **IntRect arena**. Use the **Zombie::spawn(x, y)** method, store them in an array, and ensure all are within bounds. [2]
9. (a) Assume that **background.sheet.png** is the sprite sheet given to you as shown in figure. Write SFML-C++ code snippet to draw one tile of grass onto the screen at position $(100, 100)$. Assume each tile in the sprite sheet is of size 50×50 . [2]



- (b) Assume the zombie's current position is (x_2, y_2) and the player's position is (x_1, y_1) . Write a code snippet to generate a bullet from the player position and move it towards the zombie. Assume the bullet is a rectangle of size 2×2 . [2]
- (c) Write a C++ SFML code snippet to resize an arena that uses **VertexArray** of **sf::Quads** from a 500×500 to a 600×600 arena, maintaining 50×50 tile size. [2]
10. (a) In SFML using C++, write a program that displays the text "Press Enter to Start" centered on the screen. When the user presses the Enter key, the text should change to "Playing...", but it must remain horizontally and vertically centered. [2]
- (b) Write a C++ program where Student and Sports are base classes and the Result class is derived from both. Use appropriate constructors in each class and demonstrate constructor invocation order. [2]
- (c) Design a C++ **Complex** class with private members **real** and **imag**. Overload the '+' operator as a member function to add two Complex objects. [2]

End of Questions

Subject Learning Outcome	*Taxonomy Level	Question Number	Marks
Write, compile and execute simple C++ programs to solve computational problems.	L3, L3, L3	1a, 1b, 1c	2, 2, 2
Become familiar with basic games using SFML library	L3, L3, L3, L5, L4, L3	2a, 2b, 2c, 3a, 3b, 3c	2, 2, 2 2, 2, 2
Implement Game Mechanics and Physics.	L3, L4, L3	8a, 8b, 8c	2, 2, 2
Develop advanced game features like sprite sheets, vertex arrays, texture management, and layering views.	L3, L3, L3, L4, L4, L5	4a, 4b, 4c, 9a, 9b, 9c	2, 2, 2 2, 2, 2
Apply advanced object-oriented programming concepts like inheritance, polymorphism, and abstraction	L4, L3, L3, L3, L3, L3	5a, 5b, 5c, 10a, 10b, 10c	2, 2, 2 2, 2 2
Implement file & socket I/O, sound effects, and refine the game's HUD & UI.	L3, L3, L4, L4, L3, L3	6a, 6b, 6c, 7a, 7b, 7c	2, 2, 2 2, 2, 2

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries an equal mark.

1. (a) Write a program that determines the largest digit present in an integer entered by the user. [2]
- (b) Write a C++ program with a function that dynamically allocates an array using a pointer to store product ratings from users. The function takes the number of users as input, collects their ratings, and returns the pointer. In **main()**, display the ratings and de-allocate the memory. [2]
- (c) Write a C++ program for a billing system using an enum **ProductCategory** with values: **ELECTRONICS**, **CLOTHING**, and **GROCERIES**. Create a function that applies category-based discounts (10%, 20%, 5%) to a given price. In **main()**, take user input for category and price, then display the final amount after discount. [2]

2. (a) You are given an array of 6 sprites. Write a code snippet using the SFML library to move the active sprite sequentially from top to bottom each time the down arrow key is pressed. [2]
- (b) Write a code snippet using the SFML library to implement a time bar that visually represents a 6-second countdown. [2]
- (c) Write a code snippet using SFML to implement a scoring mechanism where the score increases by 1 each time the Up arrow key is pressed. Display the updated score on the screen using `sf::Text`. [2]
3. (a) Construct SFML-C++ statements to draw FOUR green filled circle shapes of radius 10 on the screen 1920×1080 close to 4 corners of the screen. Additionally one center stretched circle with red filled of the same radius using `sf::CircleShape` Class Reference. [2]
- (b) Write a code snippet to toggle the player sprite position between the left and right sides of the tree using the **Left** and **Right** arrow keys. Ensure the player sprite is oriented correctly to face the tree always. [2]
- (c) Write a code snippet to play the sound file **death.wav** when a collision between the player and a branch is detected. [2]
4. (a) Assume that `MyBat` class has four int type data members and two member functions, `setData()` and `getData()` with return types void. Write the code snippet to declare the said class. [2]
- (b) Write a program to design a class with private data members and public functions as necessary to draw a rectangle shape of size (10, 10) over a window of resolution 1920×1080 respectively. [2]
- (c) Design a `Bat` class with a parameterize constructor to takes two float parameters. Write a program to create a bat of size 100×5 . The constructor receives two values that represent the position of the bat on the screen. Create a object of the class and draw the object on screen. [2]
5. (a) Design a C++ class named `Temperature` that includes a private data member to store the temperature in Celsius. Implement a parameterized constructor that accepts a Fahrenheit value as input, converts it to Celsius using the formula $C = (F - 32) \times 5/9$, and stores the result in the Celsius member variable. [2]
- (b) Design a C++ class named `AreaCalculator` that includes two overloaded member functions named `calculate(...)`. The first version of `calculate(...)` should take a single parameter (side length) and return the area of a square. The second version should take two parameters (length and breadth) and return the area of a rectangle. [2]

- (c) Define a base class named `Person` that contains protected data members name and age, and public member function constructor to initialize and `display()` to print these details. Then, create a derived class `Student` that inherits `Person` publicly and includes an additional private data member `rollNumber`. Add a constructor in `Student` to initialize all data members (including those from the base class) and override the `display()` function to print name, age, and roll number. [2]
6. (a) Write a code snippet using SFML to hide the default mouse cursor and display a custom crosshair image (`'crosshair.png'`) at the current mouse position. Ensure the crosshair moves in sync with the mouse and is centered correctly by setting its origin. [2]
- (b) In a SFML-based game, two game elements are defined: `spritePlayer`, representing the player character, and `textScore`, displaying the player's score. Assume that `mainView` is used for rendering the game world, and `hudView` is used for displaying the heads-up display (HUD). Write a code snippet that draws `spritePlayer` using `mainView` and `textScore` using `hudView`. [2]
- (c) Create an enum class named `GameState` with states `GameOver` and `Playing`, represented by two background images: `"gameLanding.png"` and `"gamePlay.png"` respectively. Write a code snippet that switches between these states when the 'X' key is pressed and displays the corresponding background using SFML. [2]
7. (a) Define and implement a `spawn()` function for a `Player` class that takes an `hudView` as input. The function should initialize the player's position to the center of the arena. [2]
- (b) Define `rotatePlayer(Vector2i mousePosition)` in the `Player` class to rotate the sprite to face the mouse using `atan2` that computes the angle between the player's center and the cursor. [2]
- (c) Assume a player character is initially positioned at the center of the screen `mainView`. Write a code snippet in C++ using SFML to move the player toward the left when the D key is pressed. Ensure that as the player moves, the camera (i.e., the `View`) follows the player so that the player remains centered on the screen. Make all necessary adjustments to the view and player update logic to achieve this behavior. [2]
8. (a) Write SFML-C++ statement to compute the angle between the player location (`x1, y1`) to the BLOATER position (`x2, y2`). Additionally, set the rotation of the BLOATER zombie sprite to an angle that faces the player. [2]